Irene Hou

University of California, San Diego (UCSD)
Cognitive Science Department

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https://houirene.github.io
Google Scholar

Research Interests

Human-Centered AI, Human-Computer Interaction, participatory AI, computing pedagogy

Education

2024 – Present
La Jolla, CA

University of California, San Diego
Ph.D. in Cognitive Science — Advisors: Philip Guo, April Yi Wang (ETH Zürich)
Anticipated graduation: 06/2029

2018 – 2022
La Jolla, CA

University of California, San Diego
B.S. in Cognitive Science (Design/Interaction Spec.), Computer Science minor
cum laude

Fellowships and Awards

Apr 2026 — **Paul and Daisy Soros Fellowship for New Americans**
Awarded nationally to 30 of 3,000+ applicants in 2026 (1% selection rate)

Apr 2025 — **National Science Foundation Graduate Research Fellowship (NSF GRFP)**
Awarded nationally to 1,000 of 15,000+ applicants in 2025 (6% selection rate)

Jan 2023 — **CES 2023 Innovation Honoree**
UX Lead · FluentPet Connect

Jan 2015 — **Bronze Medalist**, National eLit Illuminary Awards
Author · *Snowspirit: The Virgo Key*

Publications

CHI 2027 *(under review)*

[10] **Irene Hou**, Xiaotian Su, and April Wang. Grading Builds Tacit Knowledge: How Generative AI Grading Risks Estranging Instructors and TAs from Their Students. *In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.*

CHI 2027 *(under review)*

[9] **Irene Hou***, Jeongeon Park*, Risa Sundu*, and Steven Dow. DreamPlace: An AI-Powered Elicitation Platform to Help Citizens Envision Community Spaces. *In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.*

CHI 2027 *(under review)*

[8] Kate Hamilton, **Irene Hou**, Dev Patel, Sheena Nnam, Hena Patel, and Stephen MacNeil. "Stuck in a Spiral": Understanding Shame and Guilt in Computing Students' AI Use. *In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '27, 2027.*

CHI 2026

[7] **Irene Hou**, Alex Qin, Lauren Cheng, and Philip Guo. Beyond the Desk: Barriers and Future Opportunities for AI to Assist Scientists in Embodied Physical Tasks. *In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '26, 2026.*

CHI 2026

[6] **Irene Hou**, Zeyu Xiong, Philip Guo, and April Wang. "Bespoke Bots": Diverse Instructor Needs for Customizing Generative AI Classroom Chatbots. *In ACM CHI Conference on Human Factors in Computing Systems, ACM CHI '26, 2026.*

CI 2025 — [5] **Irene Hou***, Jeongeon Park*, Risa Sundu*, and Steven Dow. Value-Centered Framing Supports Inclusive Idea Convergence in Participatory Civic Design. *In ACM Collective Intelligence, ACM CI '25, 2025.*

ITiCSE 2025 — [4] **Irene Hou**, Kate Hamilton, Owen Man, Srishty Muthusekaran, Jeffin Johnykutty, Leili Zadeh, and Stephen MacNeil. "All Roads Lead to ChatGPT": How Generative AI is Eroding Social Interactions and Student Learning Communities. *In Innovation and Technology in Computer Science Education, ACM ITiCSE '25, 2025.*

ACE 2025 — [3] Sebastian Gutierrez, **Irene Hou**, Jihye Lee, Kenneth Angelikas, Owen Man, Sophia Mettille, James Prather, Paul Denny, and Stephen MacNeil. Seeing the Forest and the Trees: Solving Visual Graph and Tree Based Data Structure Problems using Large Multimodal Models. *In Australasian Computing Education Conference, ACM ACE '25, 2025.*

ACE 2024 — [2] **Irene Hou**, Sophia Mettille, Owen Man, Zhuo Li, Cynthia Zastudil, and Stephen MacNeil. The Effects of Generative AI on Introductory Students' Help Seeking Preferences. *In Australasian Computing Education Conference, ACM ACE '24, 2024.*

ACE 2024 — [1] **Irene Hou**, Owen Man, Sophia Mettille, Sebastian Gutierrez, Kenneth Angelikas, and Stephen MacNeil. More Robots are Coming: Large Multimodal Models (ChatGPT) can Solve Visually Diverse Images of Parsons Problems. *In Australasian Computing Education Conference, ACM ACE '24, 2024.*

(denotes co-first author)*

Posters, Extended Abstracts, Workshop Papers

SIGCSE 2025 — [p.1] **Irene Hou**, Hannah V. Nguyen, Owen Man, and Stephen MacNeil. The Evolving Usage of Generative AI by Computing Students. *In Technical Symposium on Computing Science Education, ACM SIGCSE '25, 2025. Poster.*

Research Experience

Summer 2026 — *Zürich, CH*
PEACH Lab, ETH Zürich · Visiting Researcher · *Advised by April Wang*
· AI-assisted grading and instructor tacit knowledge [10]

2024 – Present — *La Jolla, USA*
Design Lab, UC San Diego · Graduate Researcher · *Advised by Philip Guo*
· AI for scientists' embodied lab and field work [7]

2024 – Present — *La Jolla, USA*
ProtoLab, UC San Diego · Graduate Researcher · *Advised by Steven Dow*
· Participatory civic design and AI-powered community elicitation [5, 9]

Summer 2025 — *Zürich, CH*
PEACH Lab, ETH Zürich · Visiting Researcher · *Advised by April Wang*
· Instructor needs for customizing classroom AI chatbots [6]

2023 – 2025 — *Philadelphia, USA*
Temple HCI Lab, Temple University · Research Lead · *Advised by Stephen MacNeil*
· AI impact on computing students; led 5 undergraduate research teams [1–4, 8]

2021 – 2022 — *La Jolla, USA*
ProtoLab, UC San Diego · Undergraduate Researcher
· Designed for CoBoards, a digital whiteboard system

Spring 2021 — **Comparative Cognition Lab, UC San Diego** · Undergraduate Research Assistant
La Jolla, USA — · Designed interface for a video-analysis behavioral encoding tool

Additional Work Experience

2021 – 2023 — **FluentPet** · UX Lead
San Jose, USA
· Led designers and developers in the end-to-end design, build, and launch of FluentPet Connect, connected-hardware app and product (CES 2023 Innovation Honoree)
· Launch achieved 72% DAU and 78.7% WAU, with 58k+ weekly interactions in first month

Summer 2020 — **SONY** · Curriculum Designer
San Diego, USA
· Designed curriculum for SONY's KOOV (edTech robotics), used by 10+ instructional teams

Summer 2017 — **Booknection Creative Writing Summer Camp** · Founder
San Diego, USA
· Founded a creative science-fiction writing camp for teen writers
· Designed original curriculum, pitched to local academies, secured funding, instructed classes

2013 – 2018 — **Snowspirit: The Virgo Key** · Author
San Diego, USA
· Published an award-winning science-fiction/fantasy novel
· Fundraised, designed PR campaigns, organized launch events with 2,000+ attendees

Invited Talks and Media

Apr 2026 — **Design@Large, UC San Diego Design Lab** · Invited Discussant
Session: Learning in the Age of GenAI: Practice, Policy, and Pedagogy

Apr 2026 — **Center for Teaching, Learning and Outreach**, California Institute of Technology
Talk: How GenAI is Eroding Social Interactions and Student Learning Communities

Jan 2026 — **The CS-Ed Podcast** · Guest
Episode S4xE14: GenAI's Impact on Student Help Seeking and More

Jun 2025 — **Responsible Tech Institute Student Network**, University of Oxford
Talk: How GenAI is Eroding Social Interactions and Student Learning Communities

May 2024 — **Raspberry Pi Foundation**, United Kingdom (remote)
Seminar: Generative AI is changing undergraduate education and undergraduate research

Teaching

2021 – 2025 — **Teaching Assistant** · UC San Diego, Department of Cognitive Science
La Jolla, CA
COGS 124: HCI Technical Systems Research (Spring 2025)
COGS 127: Data-Driven UX/Product Design (Winter 2025, Winter 2021)
COGS 13: Field Methods: Studying Cognition in the Wild (Winter 2022)

2017 – 2022 — **Instructor** · K–12 Education
San Diego, CA
Grace Academy (2019–2022): Robotics, Python, Physics, Computational Thinking
Starton Tech EDU (2018–2020): Python, Computational Thinking, KOOV Robotics
All-Star Academy (2017–2019): Creative Writing, Introduction to Programming

Guest Lectures

May 2026 — COGS 10: Cognitive Consequences of Technology, UC San Diego
Lecture: Social Impacts of AI in Education

May 2025	Canyon Crest Academy, San Diego Seminar: Introduction to HCI and Exploring HCI Careers
Feb 2025	COGS 124: HCI Technical Systems Research, UC San Diego Lecture: Fairness, Accountability, Trust, Ethics in Human-Centered AI

Academic Service

Reviewer	CHI ('24, '25, '26), DIS '26, SIGCSE '26
Student Volunteer	SIGCSE '25

Leadership and Outreach

2025 – Present	**Vice President, Graduate Women in Computing @ UC San Diego** Organized and launched the Undergraduate Research Night series (3 events, 2025–2026)
Winter 2025	**Outreach Co-Chair, Graduate Women in Computing @ UC San Diego**

Research Mentorship

Undergraduates I have directly supervised in research or mentored. *(†denotes community college student)*

Owen Man†, De Anza College [1–4, p.1] → MS, CSU San Marcos → Microsoft
Kenneth Angelikas, Temple University [2–3] → Microsoft
Kate Hamilton, Temple University [4, 8] → Comcast
Ayah Ibrahim, UC San Diego → PhD, UC Santa Cruz
Olivia Lam†, Palomar College → UCLA (Data Science)
Brandson Trinh†, De Anza College → UC San Diego (Data Science)
Zoe Kathari, UC San Diego (Honors Thesis, Cognitive Science)
Alex Qin, UC San Diego [7]
Lauren Cheng, UC San Diego [7]
Sophia Mettille, Temple University [1–2]
Sebastian Gutierrez, Temple University [2–3]
Zhuo Li, Temple University [1]
Srishty Muthusekaran, Temple University [4]
Jeffin Johnykutty, Temple University [4]
Leili Massoum Zadeh, Temple University [4]
Hannah Vy Nguyen, Temple University [p.1]
Ben Nguyen, Temple University
Alexander Yu, UC San Diego
Katie Liu, UC San Diego